URBAN CLIMATE CONFERENCE 2026 · LISBON · POSTER P-14

Cool streets

Three decades of urban heat islands in twelve European cities, seen from orbit



Maya Okafor[1] **Tomás Rivera**[1] **Lena Hartmann**[2]

[1] Institute for Urban Climate, Northfield University [2] Centre for Earth Observation, Halden

UNIVERSITY LOGO

PARTNER LOGO

01 BACKGROUND

Why streets get hot

Built surfaces store heat by day and release it slowly at night [1]. The street, not the weather station, is where people feel it.

61 672
heat-related deaths in Europe in the summer of 2022 [2]

1 in 3
Europeans live in a city of more than 100 000 people

02 DATA AND METHOD

From orbit to street

Every cloud-free summer scene since 1990 goes through the same five steps.

01 **Landsat 5, 7, 8 and 9**
1 264 summer scenes, 1990 to 2024

02 **Cloud and quality mask**
QA band plus a buffer around every cloud

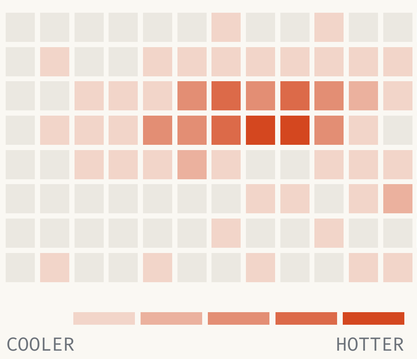
03 **Surface temperature**
Single-channel retrieval [3]

04 **100 m street grid**
4 812 neighbourhoods, each with canopy and land cover

05 **Mixed-effects model**
Random intercept per city

$$T_s = \frac{T_B}{1 + (\lambda\, T_B/\rho)\, \ln \varepsilon} \qquad \Delta T = \overline{T_s}^{\,\text{urban}} - \overline{T_s}^{\,\text{rural}}$$

Brightness temperature T_B at 10.9 µm, $\rho = hc/k_B$, and emissivity ε from NDVI. The heat island ΔT is computed scene by scene.



Land surface temperature, Madrid centre, 3 August 2024, in five classes.

03 KEY FINDING

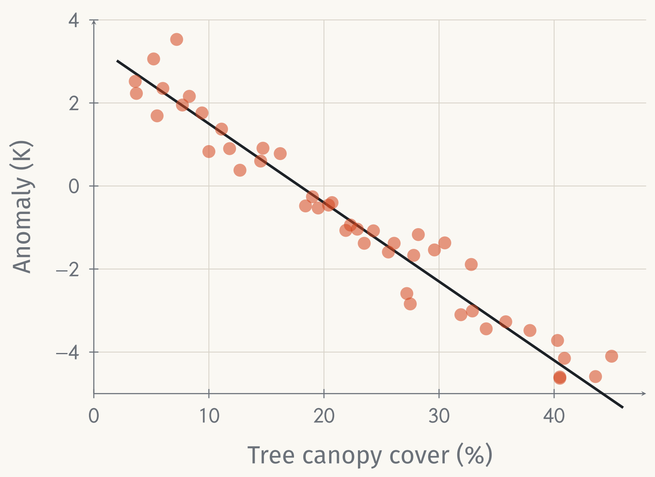
Trees beat every lever

PER TEN POINTS OF TREE CANOPY

−1.9 K

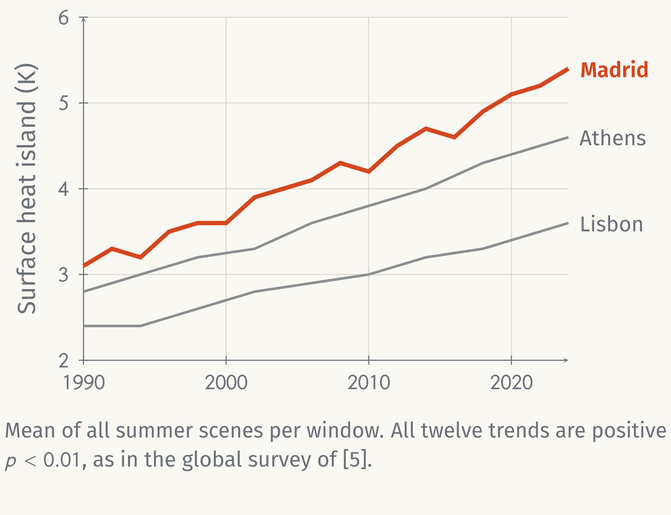
A street with ten more points of canopy is almost two degrees cooler, more than any other factor a city controls.

Canopy against surface anomaly



Each dot bins 100 neighbourhoods from all twelve cities; the line is the pooled fit, in line with [4].

Heat island by year

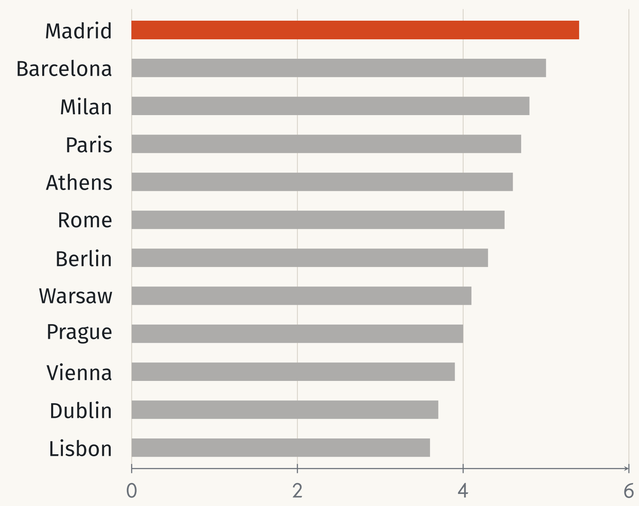


Mean of all summer scenes per window. All twelve trends are positive at $p < 0.01$, as in the global survey of [5].

04 RESULTS

Twelve cities, one pattern

Summer heat island, 2024



Urban mean minus rural-ring mean, all scenes of the 2024 summer.

What drives the anomaly

Driver	Effect (K)	Share
Impervious surface, +10 pp	+1.42	38%
Tree canopy, +10 pp	−1.87	29%
Roof albedo, +0.1	−0.94	12%
Building height, +5 m	+0.36	9%
Distance to water, +1 km	+0.21	5%

Share of explained variance; $n = 4\,812$ neighbourhoods, $R^2 = 0.71$.

05 IMPLICATIONS

What a city can do

01 **Plant where it is hottest**
The hottest 10% of blocks hold 31% of residents over 65.

02 **Lighten the roofs**
Albedo +0.1 recovers half the effect of canopy, at a fraction of the cost.

03 **Measure from orbit**
Public Landsat data lets any city track its heat island every summer.

REFERENCES

[1] Timothy R. Oke. “The energetic basis of the urban heat island”. In: *Quarterly Journal of the Royal Meteorological Society* 108.455 (1982), pp. 1–24. DOI: 10.1002/qj.49710845502.

[2] Joan Ballester et al. “Heat-related mortality in Europe during the summer of 2022”. In: *Nature Medicine* 29 (2023), pp. 1857–1866. DOI: 10.1038/s41591-023-02419-z.

[3] Juan C. Jiménez-Muñoz and José A. Sobrino. “A generalized single-channel method for retrieving land surface temperature from remote sensing data”. In: *Journal of Geophysical Research: Atmospheres* 108.D22 (2003). DOI: 10.1029/2003JD003480.

[4] Jonas Schwaab et al. “The role of urban trees in reducing land surface temperatures in European cities”. In: *Nature Communications* 12 (2021), p. 6763. DOI: 10.1038/s41467-021-26768-w.

[5] Shushi Peng et al. “Surface urban heat island across 419 global big cities”. In: *Environmental Science & Technology* 46.2 (2012), pp. 696–703. DOI: 10.1021/es2030438.

DATA AND CODE

https://example.org/cool-streets

CONTACT

m.okafor@example.org